WORKED SOLUTION

STEP III 1993, Question 5

Pure • Groups and Quaternions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 6.7, *The Quaternion Group*: https://beyondstep.keithmcnulty.org/ch06/quaternions.html

The key idea

The operation looks arbitrary, but the four products in the first part show its structure: pairs of the form $(z,0)$ multiply like ordinary complex numbers, and the second slot behaves like a new symbol j with $j^2=-1$, which conjugates anything it moves past. Once you have those four products, the eight elements of S_1 are just $(u,0)$ and $(0,u)$ with $u\in\{1,i,-1,-i\}$. So closure, inverses and orders come from a few general lines, with no need for a 64-entry table.

Solution

Associativity and commutativity

Write $z=(z_1,z_2)$, $w=(w_1,w_2)$, $u=(u_1,u_2)$. We use $(ab)^*=a^*b^*$, $(a\pm b)^*=a^*\pm b^*$ and $a^{**}=a$ throughout. First,

$$z\circ w=(z_1w_1-z_2w_2^*,\ z_1w_2+z_2w_1^*),$$

so the two components of $(z\circ w)\circ u$ are

$$\begin{aligned}(z_1w_1-z_2w_2^*)u_1-(z_1w_2+z_2w_1^*)u_2^* &= z_1w_1u_1-z_1w_2u_2^*-z_2w_1^*u_2^*-z_2w_2^*u_1,\\ (z_1w_1-z_2w_2^*)u_2+(z_1w_2+z_2w_1^*)u_1^* &= z_1w_1u_2+z_1w_2u_1^*+z_2w_1^*u_1^*-z_2w_2^*u_2.\end{aligned}$$

On the other side, $w\circ u=(w_1u_1-w_2u_2^*,\ w_1u_2+w_2u_1^*)$, so the two components of $z\circ(w\circ u)$ are

$$\begin{aligned}z_1(w_1u_1-w_2u_2^*)-z_2(w_1u_2+w_2u_1^*)^* &= z_1w_1u_1-z_1w_2u_2^*-z_2w_1^*u_2^*-z_2w_2^*u_1,\\ z_1(w_1u_2+w_2u_1^*)+z_2(w_1u_1-w_2u_2^*)^* &= z_1w_1u_2+z_1w_2u_1^*+z_2w_1^*u_1^*-z_2w_2^*u_2.\end{aligned}$$

The right-hand sides agree term by term, so $\circ$ is associative.

It is not commutative. For example,

$$(0,1)\circ(i,0) = (0-0,\ 0+1\cdot i^*) = (0,-i), \qquad (i,0)\circ(0,1) = (0-0,\ i\cdot 1+0) = (0,i).$$

The four products

Putting zeros into the definition,

$$(z,0)\circ(w,0) = (zw,\ 0), \qquad (z,0)\circ(0,w) = (0,\ zw),$$
$$(0,z)\circ(w,0) = (0,\ zw^*), \qquad (0,z)\circ(0,w) = (-zw^*,\ 0).$$

The set S_1

Let $U = \{1, i, -1, -i\}$. Then

$$S_1 = \{(u,0) : u \in U\} \cup \{(0,u) : u \in U\},$$

with A, C, E, G of the first kind and B, D, F, H of the second.

Closure. The set U is closed under multiplication, under complex conjugation ($i^* = -i$) and under negation. By the four products, the product of two elements of S_1 is $(zw,0)$, $(0,zw)$, $(0,zw^*)$ or $(-zw^*,0)$ with $z,w \in U$. Each of zw and $\pm zw^*$ lies in U, so every product lies in S_1.

Identity. For any $(w_1,w_2) \in S$,

$$(1,0)\circ(w_1,w_2) = (w_1,\ w_2), \qquad (w_1,w_2)\circ(1,0) = (w_1,\ w_2\cdot 1^*) = (w_1,w_2),$$

so $A = (1,0)$ is a two-sided identity, and it lies in S_1.

Inverses and orders. For elements of the first kind, $(z,0)\circ(w,0) = (zw,0)$, so they behave exactly like the complex numbers in U. Hence A has order 1 and is its own inverse, $E = (-1,0)$ has order 2 and is its own inverse, and $C = (i,0)$ and $G = (-i,0)$ each have order 4 and are inverses of each other.

> **Ask yourself** What is the square of an element of the second kind?

By the last of the four products, for $u \in U$,

$$(0,u)\circ(0,u) = (-uu^*,\ 0) = (-|u|^2,\ 0) = (-1,0) = E.$$

So each of B, D, F, H squares to E. Since E has order 2, each of these has order 4. For the inverse, $(0,u)\circ(0,w) = (-uw^*,0)$ equals $(1,0)$ exactly when $w^* = -1/u = -u^*$ (using $uu^* = 1$), that is, when $w = -u$. Checking the other order, $(0,-u)\circ(0,u) = (uu^*,0) = (1,0)$ too. So the inverse of $(0,u)$ is $(0,-u)$.

In summary:

element	A	B	C	D	E	F	G	H
inverse	A	F	G	H	E	B	C	D
order	1	4	4	4	2	4	4	4

Group. We now have all four axioms. S_1 is closed under $\circ$. The operation is associative on all of S, so in particular on S_1. $A \in S_1$ is an identity, and every element has an inverse in S_1. So S_1 is a group under $\circ$, of order 8. It is non-abelian: $B \circ C = H$ but $C \circ B = D$.

The subgroup and the matrix

Using the products above, the powers of $B = (0,1)$ are

$$B^2 = E, \qquad B^3 = E \circ B = (-1,0) \circ (0,1) = (0,-1) = F, \qquad B^4 = E \circ E = A.$$

So $\{A, B, E, F\} = \{B^0, B^1, B^2, B^3\}$ is the cyclic subgroup generated by B. It is closed, because $B^m \circ B^n = B^{m+n}$ and exponents can be reduced mod 4, and it contains A and the inverse B^{4-k} of each B^k. So it is a subgroup of order 4.

Let $M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then

$$M^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I, \qquad M^3 = -M, \qquad M^4 = I,$$

so the group generated by M is $\{I, M, -I, -M\}$, also cyclic of order 4. Any two cyclic groups of the same order are isomorphic. Here the map

$$A \mapsto I, \quad B \mapsto M, \quad E \mapsto -I, \quad F \mapsto -M,$$

that is, $B^k \mapsto M^k$, is a bijection and satisfies $B^m \circ B^n = B^{m+n} \mapsto M^{m+n} = M^m M^n$. So the two groups *are* isomorphic.

Checks

- Order of an element divides the order of the group: all the orders $1, 2, 4$ divide 8, and 2 and 4 divide 4 for the subgroup.
- The complex-number view: $(z,0) \circ (w,0) = (zw,0)$, so $C \circ C = (i^2,0) = E$, exactly as $i^2 = -1$.
- A short computer check confirmed associativity on 10 000 random triples of complex pairs, built the full 8×8 table to confirm closure, the identity, the inverses and orders above and $B \circ C = H \neq D = C \circ B$, and confirmed that $B^k \mapsto M^k$ respects multiplication.

Common slips

- Forgetting the conjugates: $(0,z) \circ (w,0)$ is $(0, zw^*)$, not $(0, zw)$. This is where non-commutativity comes from.
- Showing associativity with one numerical example. An example can show that a law *fails*, but proving it needs the general expansion.

- Concluding "not isomorphic" because one group is made of pairs of complex numbers and the other of matrices. Isomorphism is about the structure of the multiplication, not what the elements are.

Beyond STEP

The operation is Hamilton's multiplication of *quaternions*. The pair (z_1, z_2) stands for $z_1 + z_2 j$, where $j^2 = -1$ and $jz = z^* j$. The group S_1 is the *quaternion group* $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. It is the smallest non-abelian group in which every subgroup is normal, and it differs from the other non-abelian group of order 8 (the symmetries of a square) by having only one element of order 2. Allenby's *Rings, Fields and Groups* covers groups of small order and the quaternions.

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